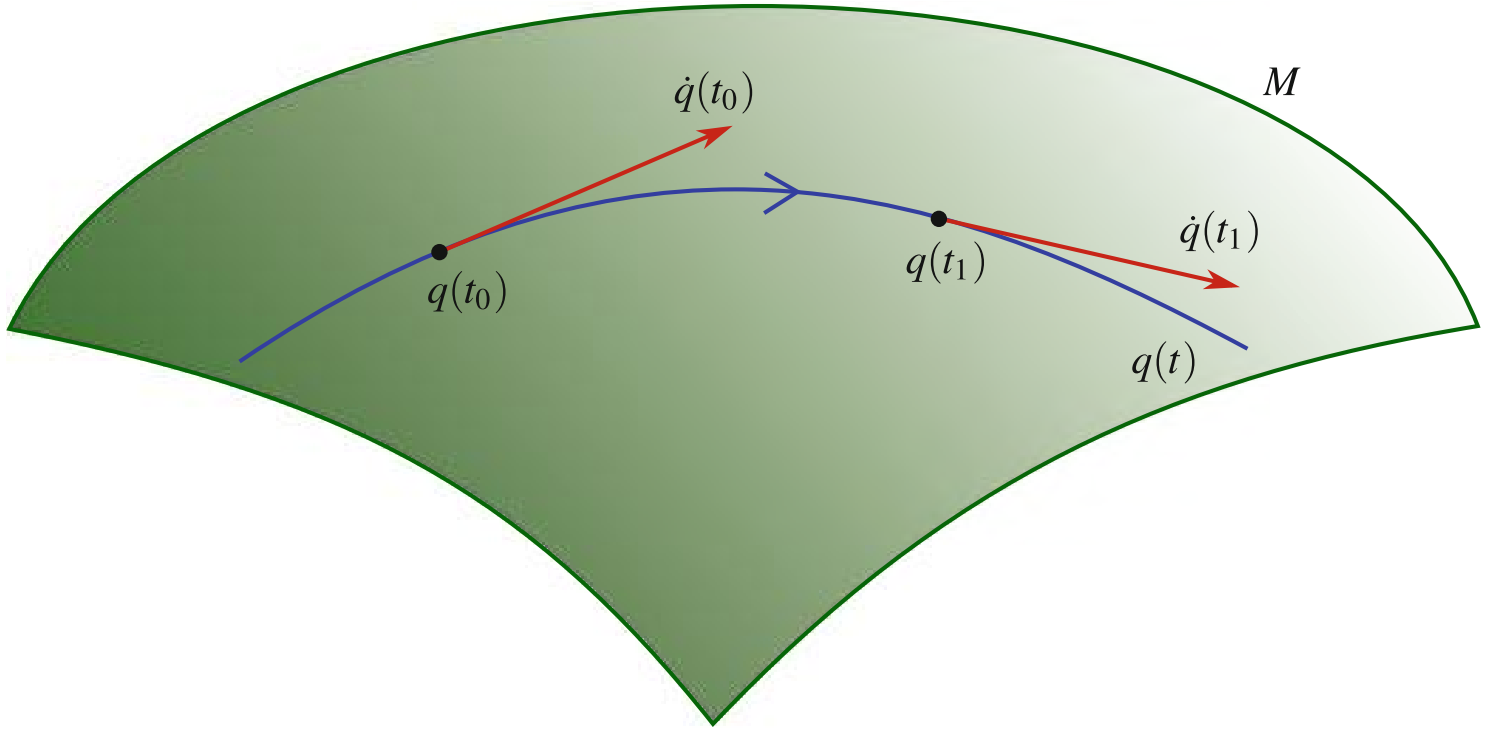
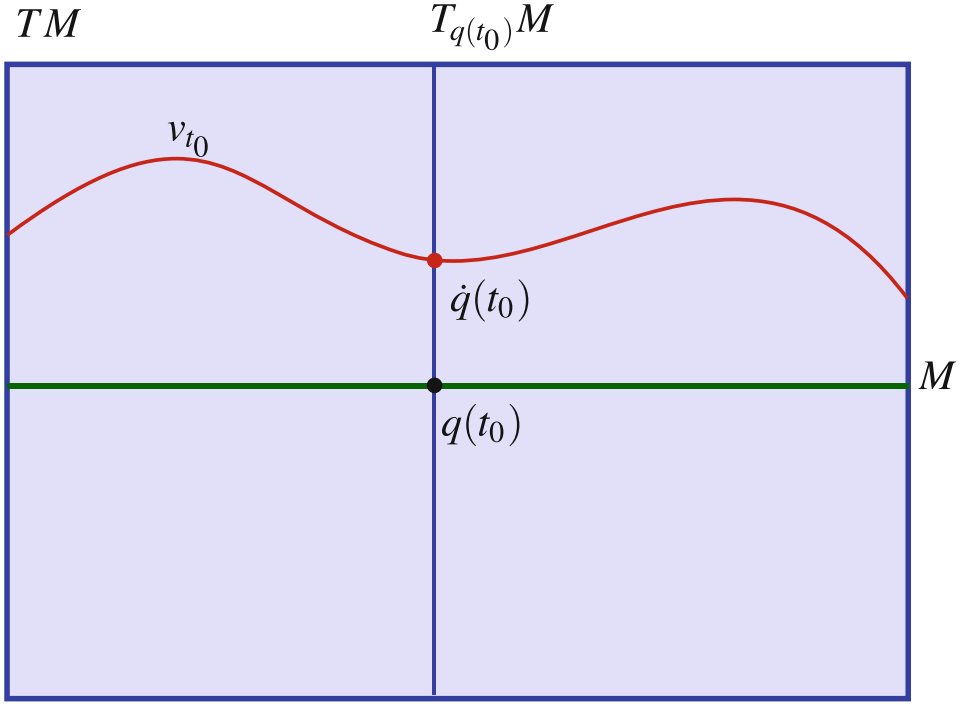
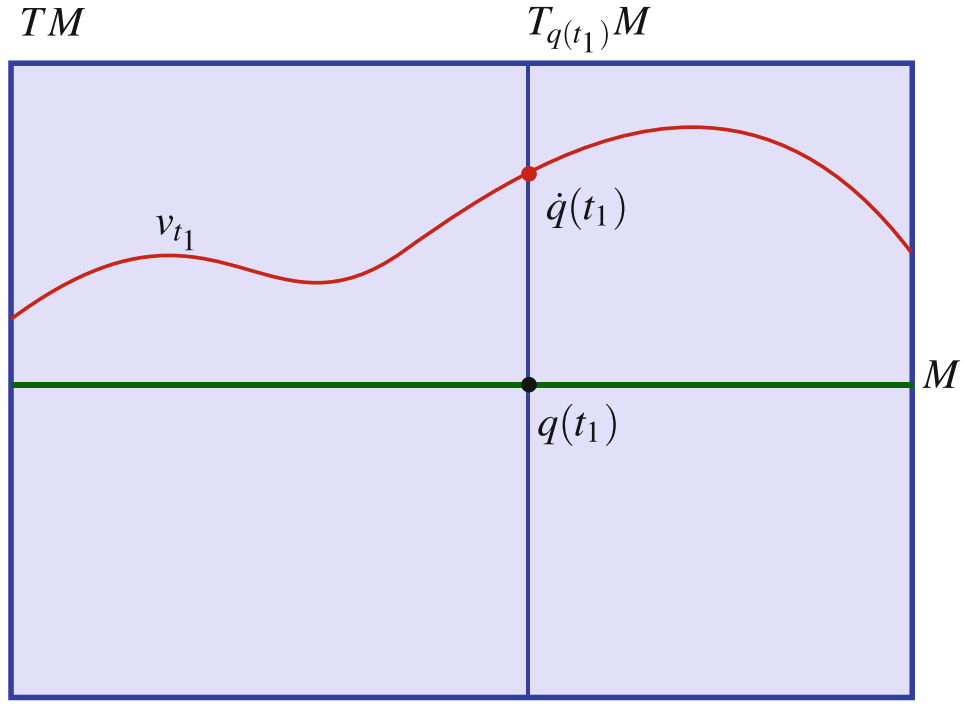

$\dot{q}(t_0)$
$M$
$q(t_0)$
$q(t_1)$
$\dot{q}(t_1)$
$q(t)$


$TM$
$T_{q(t_0)}M$
$v_{t_0}$
$\dot{q}(t_0)$
$M$
$q(t_0)$


$TM$
$T_{q(t_1)}M$
$v_{t_1}$
$\dot{q}(t_1)$
$M$
$q(t_1)$